EXCELLENCE

CLASS 11 - 2026-27

Physics | Vector 03

Vectors

Magnitude and Dot Product

By - Excellence Programme

1 October 2026

Topics to be covered

01 Magnitude from components

02 Triangle law of addition

03 Resultant formula and range

04 Equal magnitudes

05 Subtraction of vectors

06 Addition by components

07 Dot product and angle

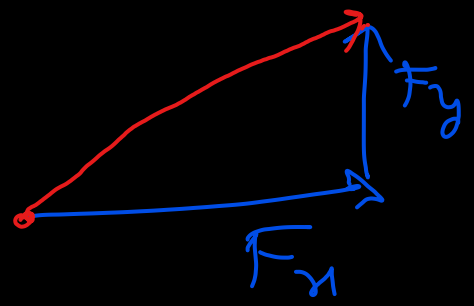
08 Unit vector

Magnitude from components

$$|F| = \sqrt{(F_x^2 + F_y^2 + F_z^2)}$$

$F_x = \vec{F}\cos\theta$

$F_y = \vec{F}\sin\theta$

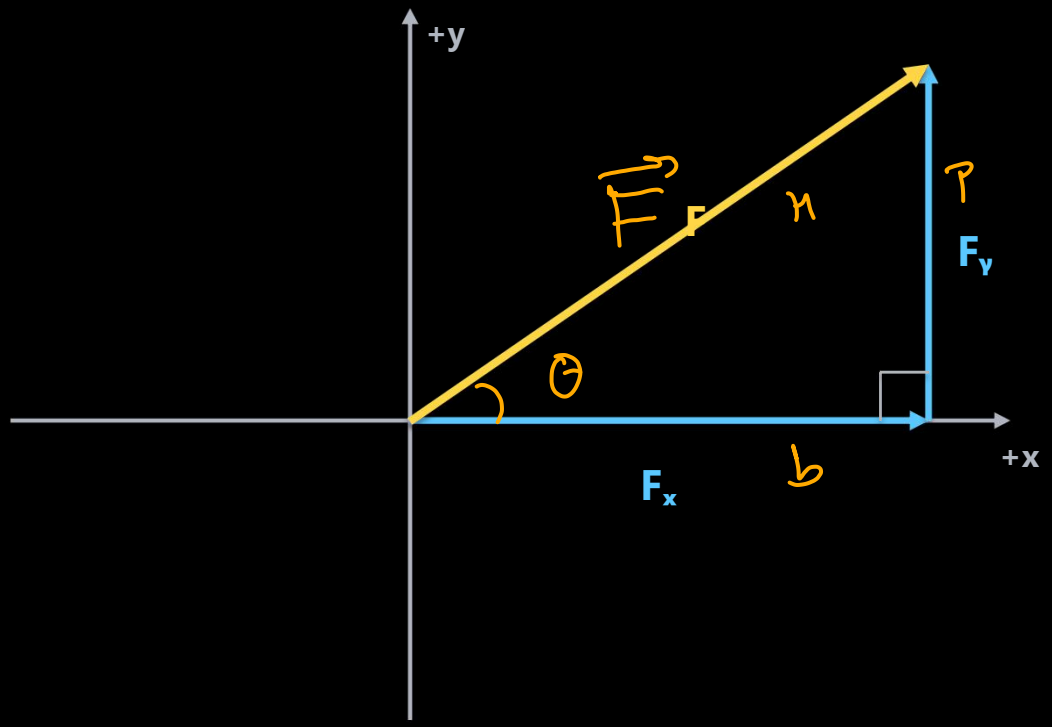


$|F|^2 = \sqrt{F_x^2 + F_y^2 + }$

Magnitude of vector in term of components

Magnitude is the length of the diagonal

Components form a right triangle. The vector is the diagonal, so Pythagoras gives its length.



$$F^2 = F_x^2 + F_y^2 \quad \text{so} \quad |F| = \sqrt{(F_x^2 + F_y^2)}$$

$h^2 = b^2 + p^2$ so $h = \sqrt{b^2 + p^2}$

With z-component

In 3D, add one more perpendicular square: $|F| = \sqrt{(F_x^2 + F_y^2 + F_z^2)}$.

Magnitude of 30 î + 40 ĵ

$\Rightarrow F_x \hat{i} + F_y \hat{j}$

Solve It!

Find the magnitude of F = 30 î + 40 ĵ.

$F_x = ?$ $30\hat{i} = 30$

$F_y = ?$ $40\hat{j} = 40$

$\not\Rightarrow$

$|F| = ? = \sqrt{30^2 + 40^2}$

$|F| = 50$

Solution: Magnitude of 30 î + 40 ĵ

Step 1: $|F| = \sqrt{(30^2 + 40^2)} = \sqrt{(900 + 1600)}$

Step 2: $\sqrt{2500} = 50$

50

Rough work / extra work

Magnitude when a component is negative

Find $|2\,\hat{i} + 2\,\hat{j}|$, $|10\,\hat{i} + 10\,\hat{j}|$ and $|3\,\hat{i} - 4\,\hat{j}|$.

$\sqrt{2^2 + 2^2}$

$= \sqrt{8}$

$10\sqrt{2}$

$\sqrt{(-3)^2 + (-4)^2}$

$\sqrt{9 + 16}$

$= \sqrt{25} = 5$

Solution: Magnitude when a component is negative

Step 1: $\sqrt{(4 + 4)} = \sqrt{8} = 2\sqrt{2}$

Step 2: $\sqrt{(100 + 100)} = 10\sqrt{2}$

Step 3: $\sqrt{(9 + 16)} = \sqrt{25} = 5$. The minus does not enter the square.

$2\sqrt{2}$, $10\sqrt{2}$, and 5.

Three components

Solve It

Find |3 î + 4 ĵ + 5 k̂| and |6 î - 8 ĵ + 10 k̂|.

$5\sqrt{2}$

$\sqrt{9+16+25}$

$= \sqrt{50}$

$5\sqrt{2}$

$\sqrt{36+64+100}$

$10\sqrt{2}$

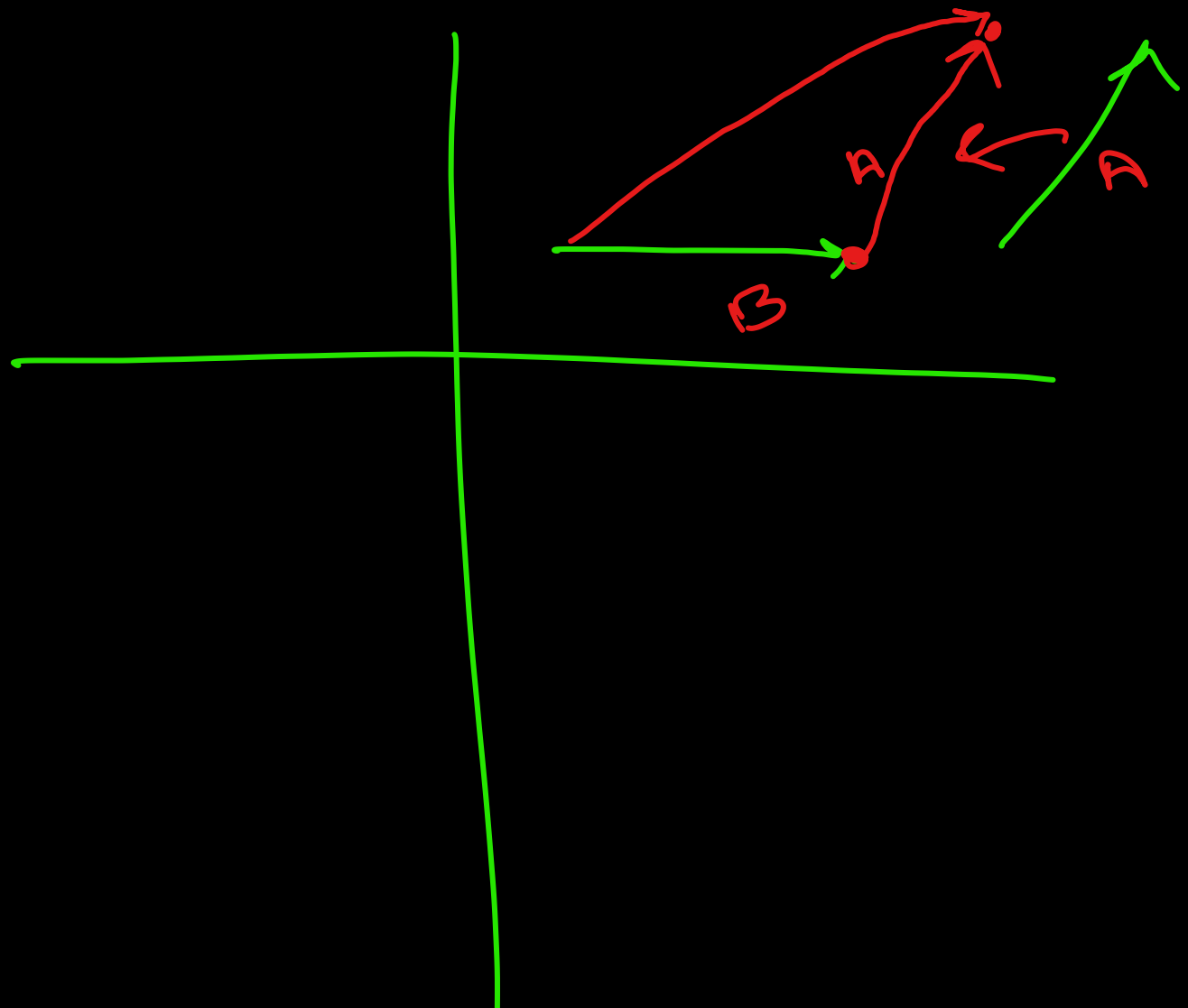
Solution: Three components

Step 1: $\sqrt{(9 + 16 + 25)} = \sqrt{50} = 5\sqrt{2}$

Step 2: $\sqrt{(36 + 64 + 100)} = \sqrt{200} = 10\sqrt{2}$

$5\sqrt{2}$ and $10\sqrt{2}$.

Rough work / extra work



Triangle law of addition

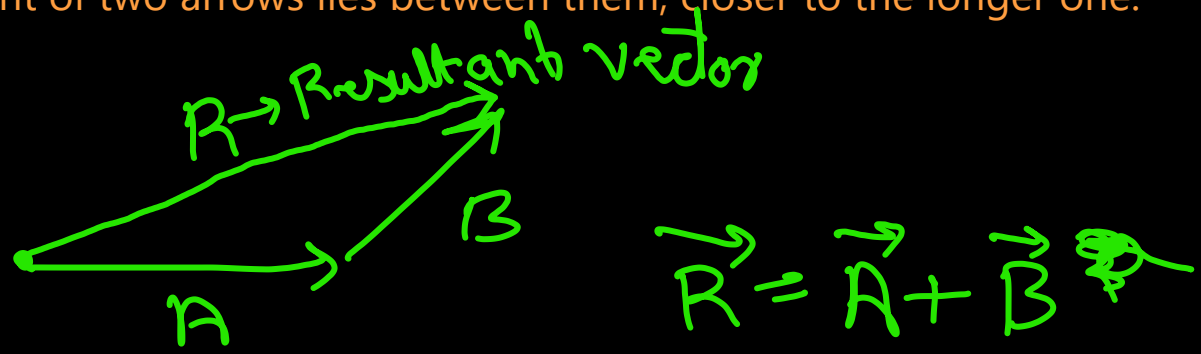
To find an angle, put tails together, or heads together.

To add, put the tail of the second arrow on the head of the first.

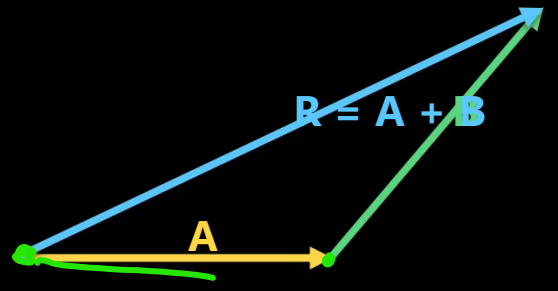
Resultant R runs from the first tail to the last head.

A + B = B + A. For three arrows the drawing order does not change R.

The resultant of two arrows lies between them, closer to the longer one.



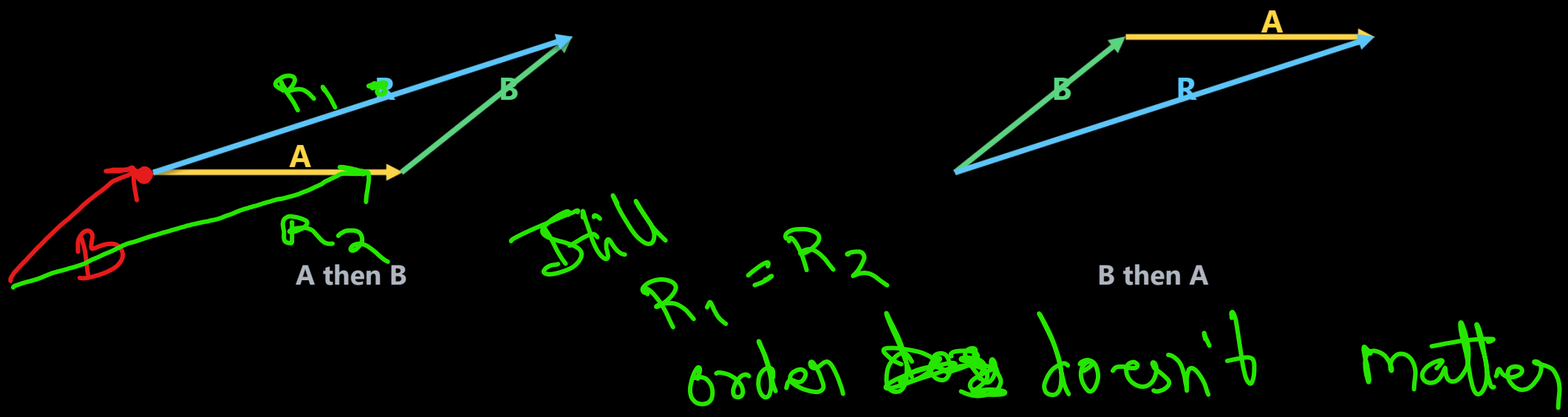
$$\vec{R} = \vec{A} + \vec{B}$$



head-to-tail

Why drawing order does not change R

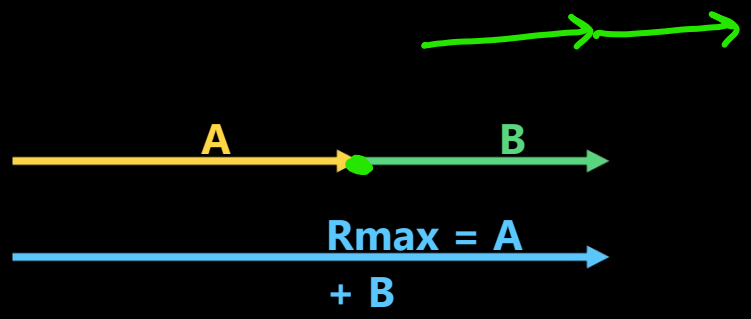
A + B and B + A reach the same final point. The path changes, but the start and finish do not.

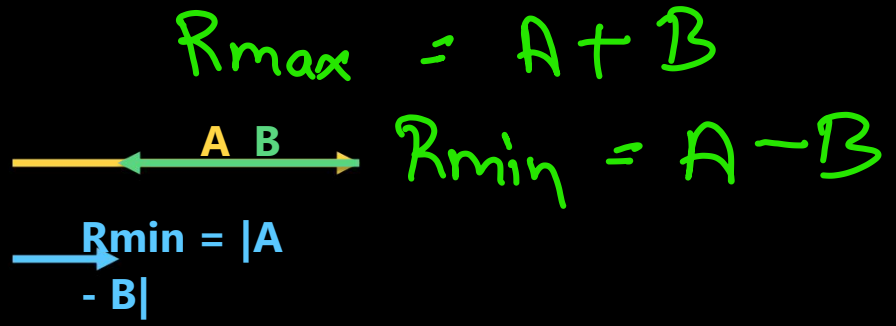


Same starting point + same ending point = same resultant.

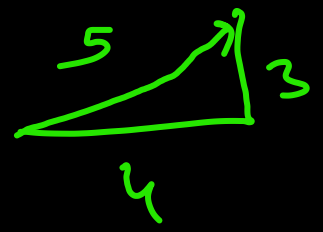
Resultant range: minimum to maximum

The largest resultant happens when both arrows pull in the same direction. The smallest happens when they oppose.



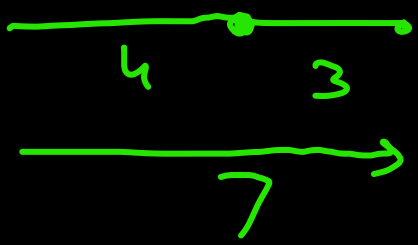


Ab same angle same
R will be ~~range~~ max

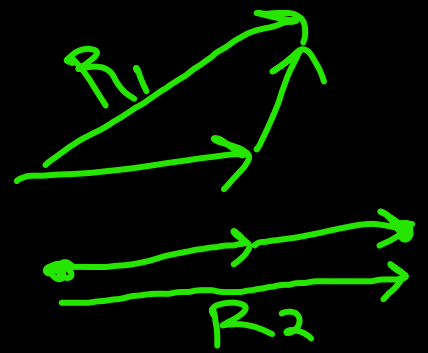


Allowed range

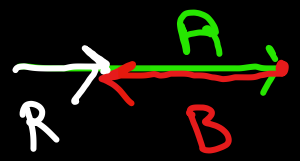
$|A - B| \le R \le A + B$



A resultant outside this range is impossible.



$R_2 = A + B$



$R = A - B$

Magnitude of the resultant

$$R = \sqrt{(A^2 + B^2 + 2\,A\,B\cos\theta)}$$

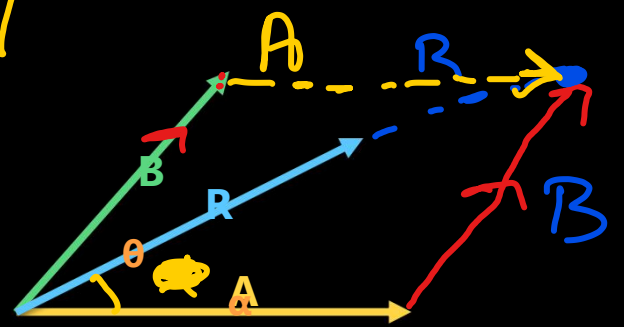
VIP

$$\tan\alpha = (B\sin\theta) / (A + B\cos\theta)$$

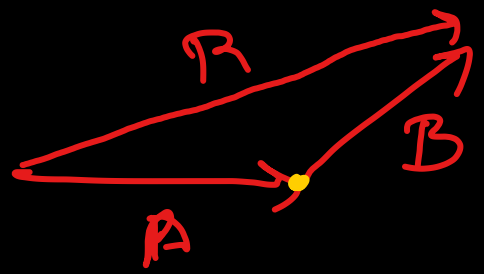
α is the angle between A and the resultant.

Pythagoras

θ = 0: R max = A + B. θ = 180: R min = |A - B|. θ = 90: R = $\sqrt{(A^2 + B^2)}$.



R is allowed only from |A - B| to A + B. Both ends are included.



$\vec{R} = \vec{A} + \vec{B}$

$|R|$

Two 4 N forces at 60°

$|R| = \sqrt{A^2 + B^2 + 2AB\cos\theta}$

|A| = 4 N, |B| = 4 N, angle = 60°. Find R.

$$R = \sqrt{16 + 16 + 2 \times 4 \times 4 \cos 60}$$

$$= \sqrt{32 + 2 \times 4 \times 4 \times \frac{1}{2}}$$

$$\sqrt{32 + 16} \Rightarrow \sqrt{48}$$

$$4\sqrt{3}$$

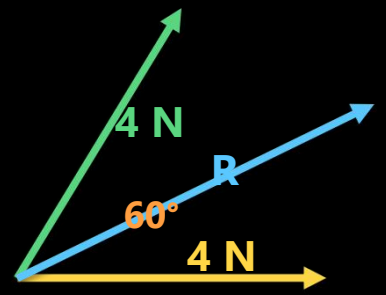
Solution: Two 4 N forces at 60°

Step 1: $R = \sqrt{(16 + 16 + 32 \times \cos 60°)}$

Step 2: $\cos 60° = 1/2$, so the middle term is 16

Step 3: $R = \sqrt{48} = 4\sqrt{3}$

$R = 4\sqrt{3}$ N.



Rough work / extra work

Magnitudes 3 and 4

Solve

Two vectors have magnitudes 3 and 4. Find R at 0°, 90° and 180°.

$\theta = 0^\circ$

$\sqrt{9+16+2\times12\times1}$

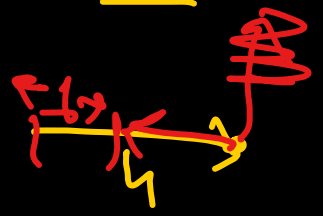
$\sqrt{49} \Rightarrow 7$

$\theta = 90^\circ$

$\sqrt{9+16+2\times12\cos90}$

$\sqrt{9+16} \Rightarrow \sqrt{25}$

$\sqrt{5}$



$\theta = 180$

$\sqrt{4^2+9^2+2\times12\cos(180}$

$\sqrt{25-2\times12\times-1}$

$= \sqrt{25-24}$

$= \sqrt{1} = 1$

Solution: Magnitudes 3 and 4

Step 1: At 0°, R = 3 + 4 = 7

Step 2: At 90°, R = √(9 + 16) = 5

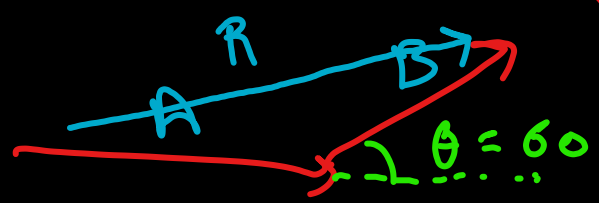
Step 3: At 180°, R = 4 - 3 = 1

7, 5 and 1.

3 and 4 at 60° and 120°

Same magnitudes 3 and 4. Find R at 60° and at 120°.

$\theta = 60$

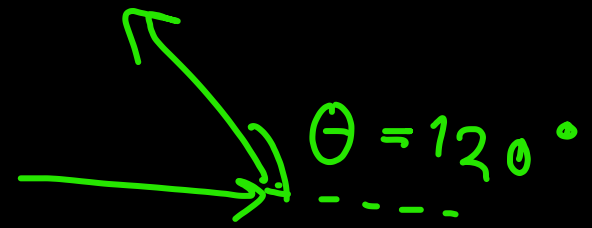


$\sqrt{9 + 16 + 2 \times 12 \times \cos 60}$

$\sqrt{24 + 2 \times 12 \times \frac{1}{2}}$

$\sqrt{37}$

$\theta = 120°$



$\sqrt{25 + 24 \times -\frac{1}{2}}$

$\sqrt{13}$

Solution: 3 and 4 at 60° and 120°

Step 1: At 60°: √(9 + 16 + 12) = √37

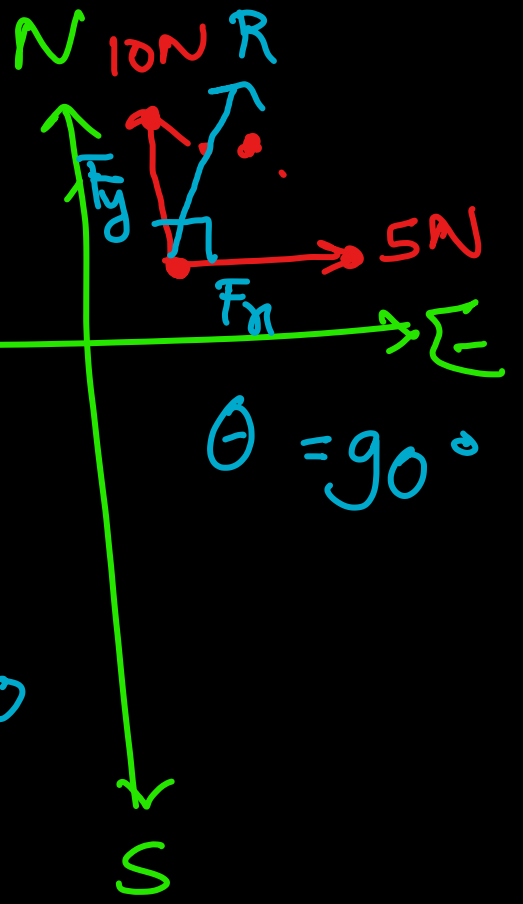
Step 2: At 120° cos = -1/2, so √(9 + 16 - 12) = √13

√37 and √13.

Rough work / extra work

5 N east and 10 N north

Two forces, 5 N due east and 10 N due north. Find the resultant.



F

F_x F_y

$R^2 = F_x + F_y$

$|R| = \sqrt{F_x^2 + F_y^2 + 2 \times F_x \times F_y \times \cos\theta}$

$\sqrt{F_x^2 + F_y^2 + 0}$ $\quad \because \cos 90 = 0$

$\sqrt{25 + 100}$

$\sqrt{125} \Rightarrow 5\sqrt{5}$

Solution: 5 N east and 10 N north

Step 1: The angle is 90°, so the cross term is 0

Step 2: $R = \sqrt{(25 + 100)} = \sqrt{125} = \sqrt{(25 \times 5)} = 5\sqrt{5}$

$5\sqrt{5}$ N.

Two 5 N forces at 120°

Solution

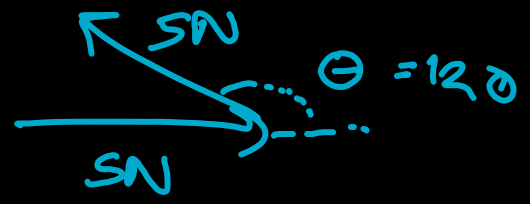
Two forces of 5 N have 120° between them. Find the net force.

$$R = 5N$$

$$R = \sqrt{5^2 + 5^2 + 2 \times 5 \times -\frac{1}{2}}$$

$$\sqrt{5^2 + 5^2 - 5^2}$$

$$\sqrt{5^2} = 5$$



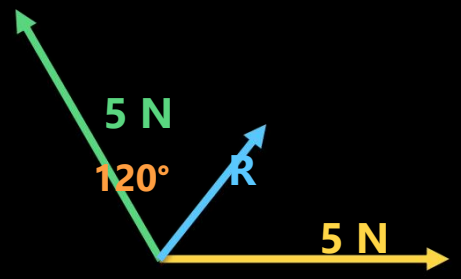


Solution: Two 5 N forces at 120°

Step 1: Use the angle between the arrows, which is 120°, not a nearby 60° mark

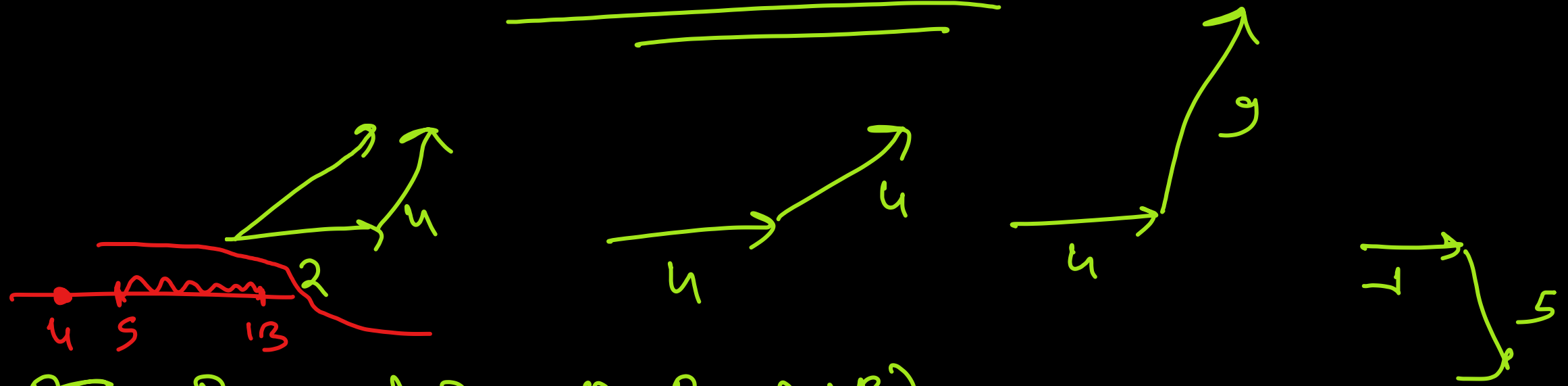
Step 2: $R = \sqrt{(25 + 25 + 50 \times (-1/2))} = \sqrt{(50 - 25)} = \sqrt{25}$

Net force = 5 N.



Which pair cannot give a resultant of 4?

Pairs: (2, 4), (4, 4), (4, 9), (1, 5). Which cannot give resultant 4?



⇒ Range of R = (A − B, A + B)

Range of (2,4) ⇒ (−2, 6) — 4 in Range

(4,4) ⇒ (0, 8) — 4 in Range

(4,9) ⇒ (−5, 13) → 4 < 5

Range of (1,5) = (−4, 6) — 4 in Range

Solution: Which pair cannot give a resultant of 4?

Step 1: (2, 4) allows 2 to 6. 4 is inside.

Step 2: (4, 4) allows 0 to 8. (1, 5) allows 4 to 6.

Step 3: (4, 9) allows 5 to 13. 4 is smaller than the minimum.

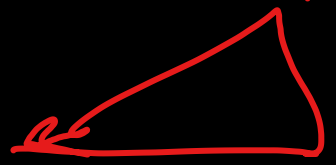
(4, 9) cannot give 4.

Rough work / extra work

Three forces, zero resultant

Solve It

Which set can have zero resultant: (2, 4, 7), (3, 1, 5), (2, 8, 11), or (3, 4, 2)?



$6 < 7$ $\quad 4 < 5$ $\quad 10 < 11$

$(7-6) \rightarrow 3$ inside the given Range

Solution: Three forces, zero resultant

Step 1: The third force must lie between |first - second| and first + second

Step 2: For 3 and 4, the range is 1 to 7. The force 2 lies inside it.

Step 3: 7 is outside 2 to 6. 5 is outside 2 to 4. 11 is outside 6 to 10.

Only (3, 4, 2) can give zero.

Equal magnitudes

If |A| = |B| = A, then R = 2 A cos(θ / 2)

VIP

Their difference has magnitude D = 2 A sin(θ / 2)

The resultant then bisects the angle. Angle between A and R is θ / 2.

θ = 0: R = 2A, D = 0. θ = 90: both are √2 A. θ = 120: R = A and D = √3 A. θ = 180: R = 0 and D = 2A.

$|A| = |B|$

$R = \sqrt{A^2 + B^2 + 2 \times A \times B \cos\theta}$

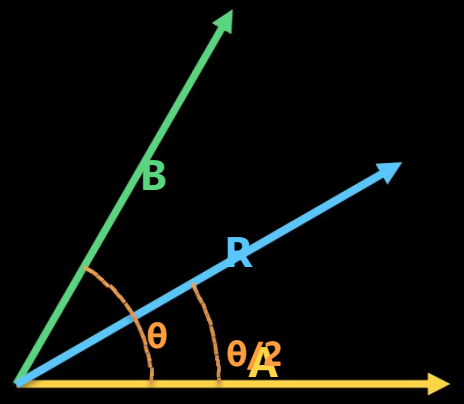
$= \sqrt{2A^2 + 2A^2\cos\theta} \Rightarrow A\sqrt{2 + 2\cos\theta}$



$\Rightarrow R = A\sqrt{2 + 2\cos\theta}$

Equal vectors: resultant bisects the angle

When |A| = |B|, the triangle is symmetric. The resultant must lie exactly in the middle.



$$R = 2A \cos(\theta / 2)$$

Fast memory

Equal vectors make an isosceles triangle, so R is the angle bisector.

Rough work / extra work

$$R = A\sqrt{2 + 2\cos\theta}$$

$$\sqrt{2(1+\cos\theta)}$$

$$\sqrt{2\left(1+\cos\left(2\cdot\frac{\theta}{2}\right)\right)}$$

$$\sqrt{2\left(2\cos^2\left(\frac{\theta}{2}\right)\right)}$$

$$\boxed{R = A\,2\cos\frac{\theta}{2}}$$

$\cos^2\theta + \sin^2\theta = 1$

$\sin^2\theta = 1 - \cos^2\theta$

$\cos 2\theta = \cos^2\theta - \sin^2\theta$

$= \cos^2\theta - 1 + \cos^2\theta$

$\cos 2\theta + 1 = 2\cos^2\theta$

$\therefore \cos 2\theta + 1 = 2\cos^2\theta$

$\therefore$ We have $\theta = \frac{\theta}{2}$

Sum and difference have the same magnitude

If |A + B| = |A - B|, what is the angle between A and B?

$A = B$,

$|A + B| = |A - B|$

$\theta = ?$

$$\frac{4AB\cos\theta}{4AB} = \frac{0}{4AB} \Rightarrow \cos\theta = 0$$

$\theta = 90^\circ$

Very good

Solution: Sum and difference have the same magnitude

Step 1: Equal lengths of sum and difference means the cross term must vanish

Step 2: $2AB\cos\theta = 0$, so $\cos\theta = 0$

$90°$.

Two unit vectors add to a unit vector

Solve it

The sum of two unit vectors is itself a unit vector. Find the magnitude of their difference.

$|\vec{A}| = 1$

$|\vec{B}| = 1$

$|\vec{R}| = 1$

$R = \sqrt{A^2 + B^2 + 2AB\cos\theta} = 1$

$= \sqrt{1 + 1 + 2\cos\theta} = 1$

Sq^r on both side

$2 + 2\cos\theta = 1^2$

$2\cos\theta = -1$

$\cos\theta = -1/2 \Rightarrow \theta = 120°$

$D = ?$

$A = B$

$D = 2AB\cos\frac{\theta}{2}$

$= 2 \times 1 \times 1 \times \frac{\sqrt{3}}{2}$

$\Rightarrow \sqrt{3}$

Solution: Two unit vectors add to a unit vector

Step 1: $R = 1$ and each magnitude is 1, so $1 = 2\cos(\theta / 2)$, so $\theta = 120°$

Step 2: $D = 2 \sin 60° = 2 \times \sqrt{3}/2 = \sqrt{3}$

Magnitude of the difference = $\sqrt{3}$.

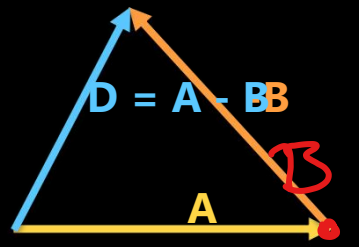
Vector subtraction

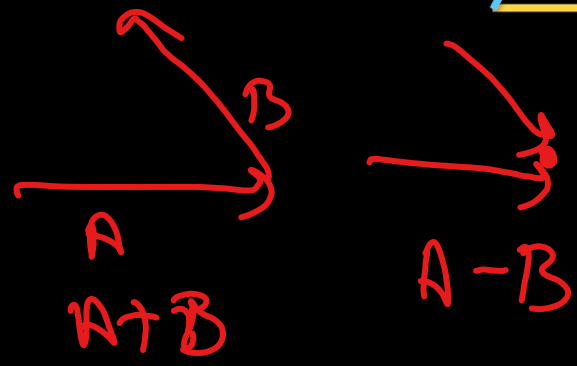
A - B = A + (-B). Reverse B, then add.

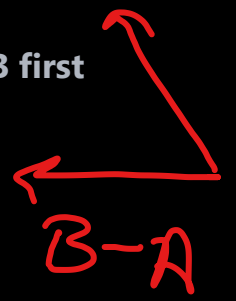
$$D = \sqrt{(A^2 + B^2 - 2AB\cos\theta)}$$

$\theta = 0$: $D = |A - B|$, the minimum. $\theta = 180$: $D = A + B$, the maximum. $\theta = 90$: $D = \sqrt{(A^2 + B^2)}$.

A + B equals B + A. A - B does not equal B - A.



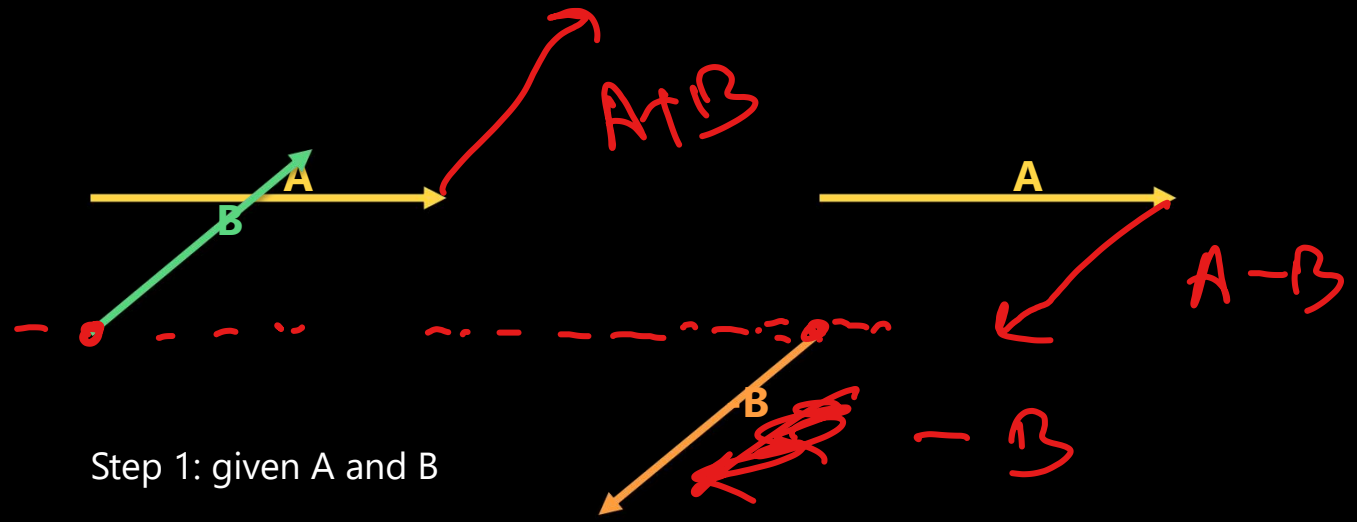




Rough work / extra work

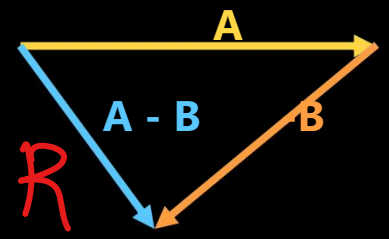
Subtraction is not a new operation

Every subtraction question is an addition question after reversing the second vector.



Step 1: given A and B

Step 2: reverse B

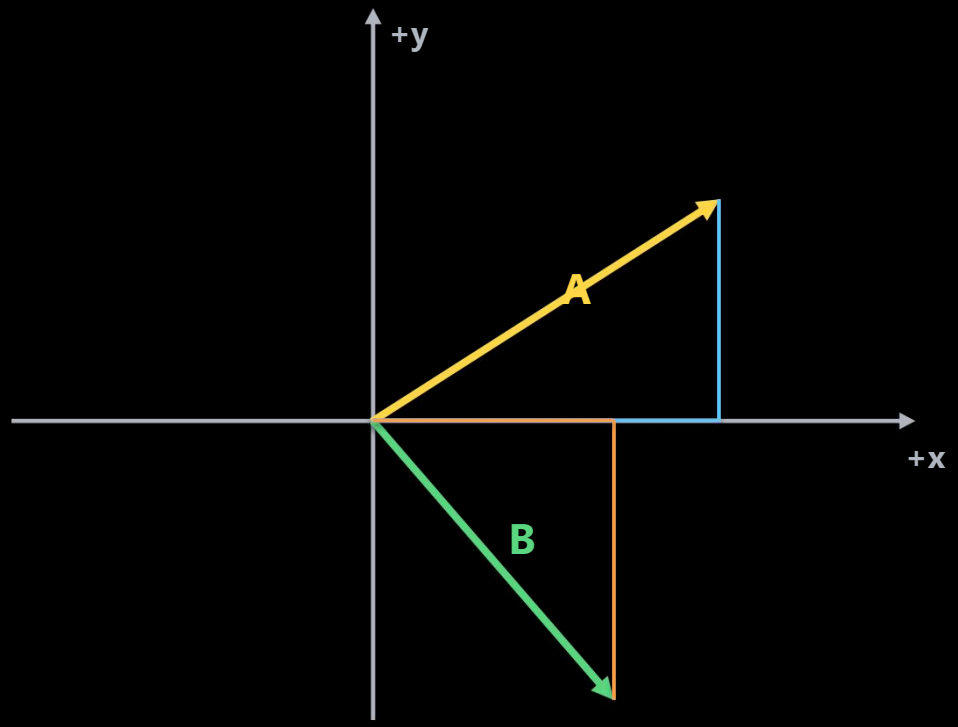


Step 3: add A + (-B)

A - B and B - A have the same length but opposite directions.

Adding by components is adding columns

The x-parts combine with x-parts. The y-parts combine with y-parts. This is the cleanest method for many vectors.



$$\mathbf{A} + \mathbf{B} = (A_x + B_x)\,\hat{\imath} + (A_y + B_y)\,\hat{\jmath}$$

Column idea

x-column decides left-right. y-column decides up-down.

3D same rule

$A + B = (A_x+B_x)\hat{\imath} + (A_y+B_y)\hat{\jmath} + (A_z+B_z)\hat{k}$

Add by components

$A = 4\hat{i} + 3\hat{j}$ and $B = 3\hat{i} - 4\hat{j}$. Find $A + B$, $A - B$, and $|A| + |B|$.